

一. 凸优化:

$$\text{形式: } \min f_0(x) \quad \text{s.t. } \underline{f_i(x) \leq 0 \quad i=1 \cdots m, \quad a_i^T x = b_i \quad i=1, \cdots p}$$

$f_0 \cdots f_m$ 凸, 线性约束

可行集凸

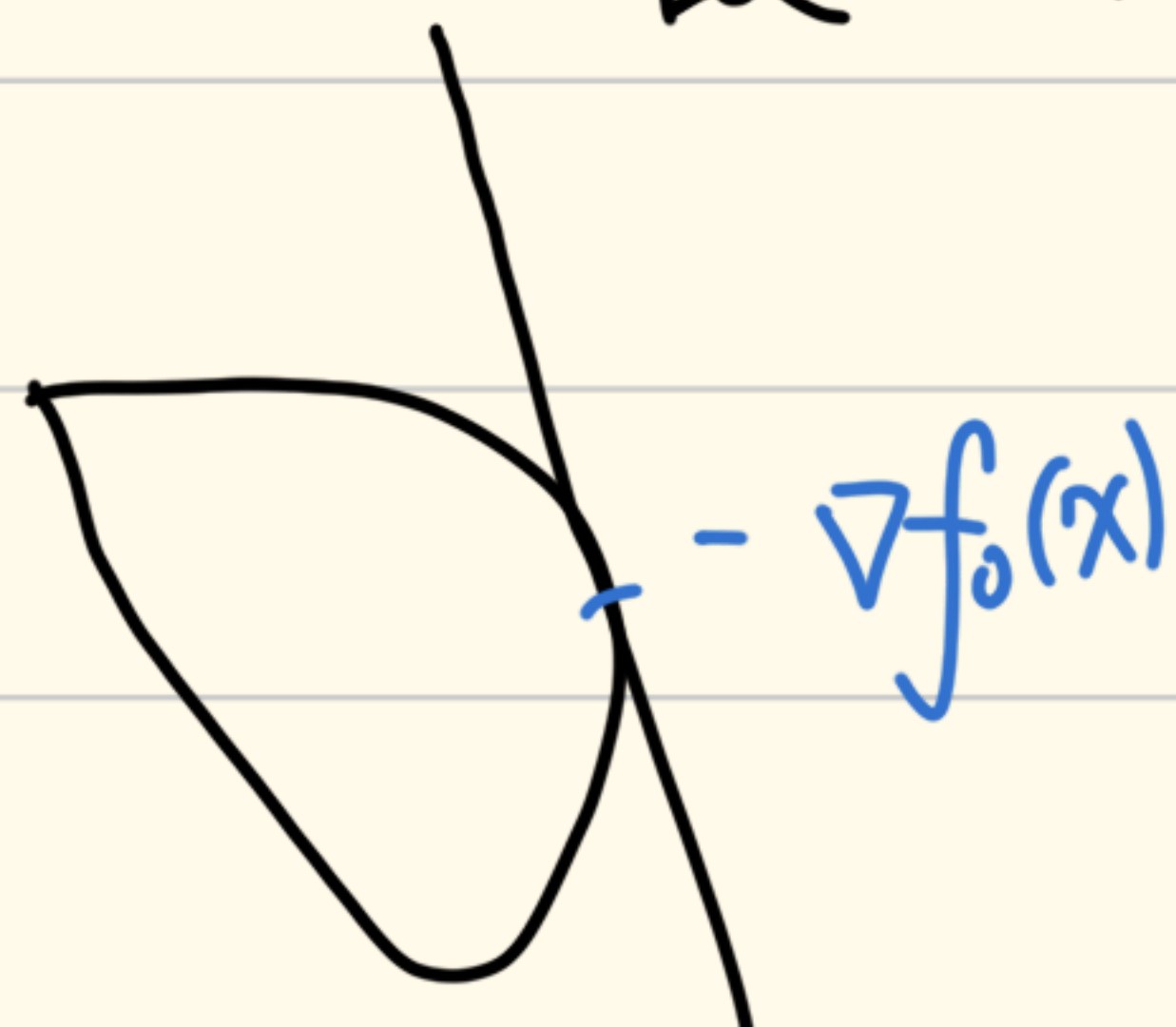
☆: 凸优化局部极小都为**全局最优** (取线性组合即可)

Thm: x 为凸优化问题 $\min_{x \in X} f_0(x)$ 最优解 iff x 可行 且 $\langle \nabla f_0(x), y-x \rangle \geq 0 \quad \forall y \in X$

$$\text{pf: } \Leftarrow: f(y) \geq f_0(x) + \langle \nabla f_0(x), y-x \rangle \geq 0$$

$$\Rightarrow: \text{若 } \langle \nabla f_0(x), y-x \rangle < 0, \text{ 记 } g(t) = f_0((1-t)x + ty), \lim_{t \rightarrow 0} g'(t) < 0, g(0) = f_0(x)$$

故 $\exists t, g(t) < g(0)$ 与最优解矛盾.



Cor: (i) 无约束优化: x 最优 iff $x \in \text{dom } f_0$, $\nabla f_0(x) = 0$

(ii) 等式约束优化: x 最优 iff 存在 v : $x \in \text{dom } f_0$, $Ax = b$, $\nabla f_0(x) + A^T v = 0$

pf: 易知 $\forall z, Az = 0 \Rightarrow \nabla f_0(x)^T z = 0 \Rightarrow \nabla f_0(x) \in \text{Im}(A^T)$

故 $\nabla f_0(x) + A^T v = 0$, 反之亦然

(iii) 非负约束优化: $\min f_0(x)$ s.t. $x \succeq 0$

x 最优 iff $x \in \text{dom } f_0$, $x \succeq 0$,

$$\begin{cases} \nabla f_0(x)_i \geq 0 & x_i = 0 \\ \nabla f_0(x)_i = 0 & x_i > 0 \end{cases}$$

pf: 取 $y = x + ke_i$ 即可

二. 线性规划

$$\text{形式: } \min_{x \in \mathbb{R}^n} C^T x \quad \text{s.t. } Ax=b, Gx \leq e \quad (A \in \mathbb{R}^{m \times n}, G \in \mathbb{R}^{p \times n}, b \in \mathbb{R}^m, e \in \mathbb{R}^p)$$

$$\text{基追踪: } \min_{x \in \mathbb{R}^n} \|x\|_1 \quad \text{s.t. } Ax=b$$

$$\text{转化: 引入 } z_i (\approx |x_i|) \quad \min_{z \in \mathbb{R}^n} \sum_{i=1}^n z_i \quad \text{s.t. } Ax=b \quad -z_i \leq x_i \leq z_i$$

$$(\text{Trick: } \min_{x, t} t \quad \text{s.t. } f(x) \leq t \Leftrightarrow \min_x f(x) \quad \text{两个问题等价})$$

$$\text{切比雪夫中心: } \mathcal{P} = \{x \mid a_i^T x \leq b_i, i=1 \cdots m\}, \text{ 最大内接球 } \mathcal{B} = \{x_c + u \mid \|u\|_2 \leq r\}$$

$$\text{此时 } a_i^T x \leq b_i \Leftrightarrow \sup_u a_i^T (x_c + u) \leq b_i \Leftrightarrow a_i^T x_c + r \|a_i\|_2 \leq b_i$$

$$\text{故 } \max_{r, x_c} r \quad \text{s.t. } a_i^T x_c + r \|a_i\|_2 \leq b_i \quad i=1 \cdots m$$

分式线性规划

$$\min f_0(x) \quad \text{s.t.} \quad Gx \leq h, Ax = b \quad f_0 = \frac{c^T x + d}{e^T x + f}, \quad \text{dom } f_0(x) = \{x \mid e^T x + f > 0\}$$

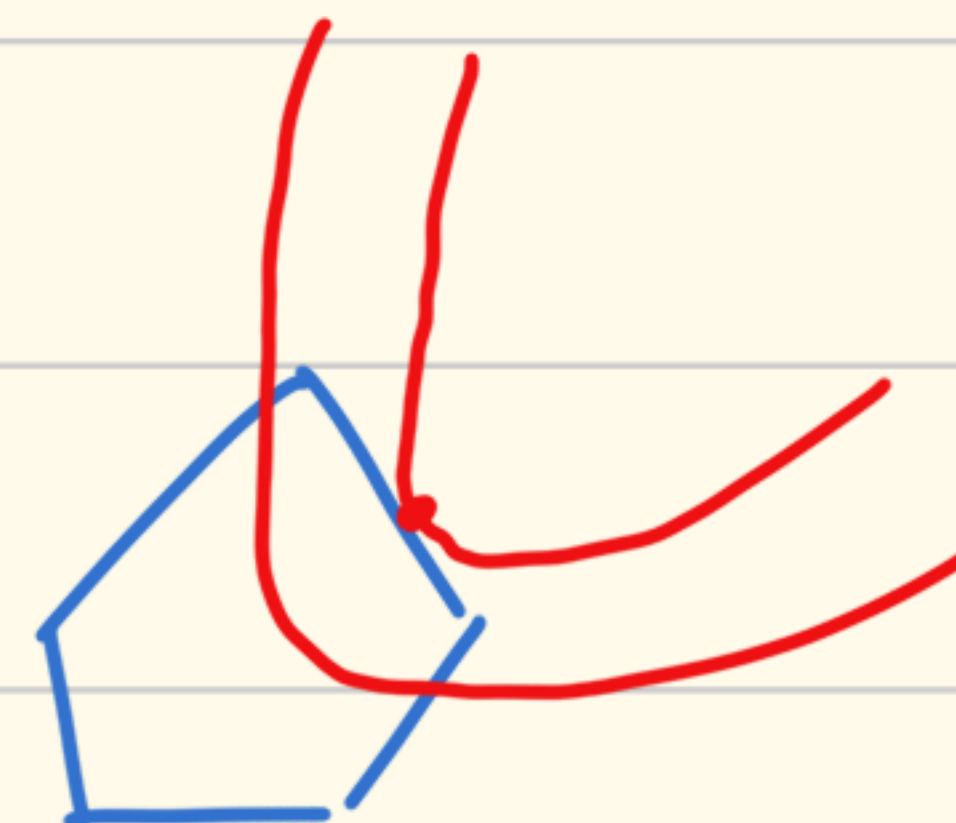
可化归为线性规划问题:

$$\min_{y, z} c^T y + dz \quad \text{s.t.} \quad Gy \leq hz, Ay = bz, e^T y + fz = 1, z \geq 0.$$

三. 二次锥规划

二次规划问题: $\min \frac{1}{2}x^T P x + q^T x + r \quad \text{s.t.} \quad Gx \leq b, Ax = b$ 其中 $P \in S_+^n$

QCQP: $\min \frac{1}{2}x^T P x + q_0^T x + r_0 \quad \text{s.t.} \quad \frac{1}{2}x^T P_i x + q_i^T x + r_i \leq 0, Ax = b$



二次锥 (SOC): $\mathcal{Q} = \{x \in \mathbb{R}^{n+1} \mid \|\bar{x}\|_2^1 \leq x_1, x = \begin{bmatrix} x_1 \\ \bar{x} \end{bmatrix}\}$

旋转锥: $\mathcal{Q} = \{x \in \mathbb{R}^{n+1} \mid \|\bar{x}\|_2^2 \leq x_1 x_2, x_1, x_2 \geq 0, x = \begin{bmatrix} x_1 \\ x_2 \\ \bar{x} \end{bmatrix}\}$

$$\text{HL: } \|\bar{x}\|^2 \leq x_1 x_2 \Leftrightarrow \left\| \begin{pmatrix} x_1 - x_2 \\ 2\bar{x} \end{pmatrix} \right\|_2 \leq |x_1 + x_2| \Leftrightarrow \left\| \begin{pmatrix} x_1 - x_2 \\ 2\bar{x} \end{pmatrix} \right\|_2 \leq \left| \frac{x_1 + x_2}{2} \right|$$

SOC P: $\min f^T x \quad \text{s.t.} \quad \|A_i x + b_i\|_2 \leq c_i^T x + d_i \quad i=1 \dots m, Fx = g$

(*) $\Leftrightarrow \begin{pmatrix} c_i^T x + d_i \\ A_i x + b_i \end{pmatrix}$ 为 SOC

二次规划 (QP) $\min q(x) = x^T Q x + a^T x + \beta, (Q \succ 0, Q^T = Q)$

s.t. $Ax = b, x \geq 0$

由 $q(x) = \|Q^{\frac{1}{2}}x + \frac{1}{2}Q^{-\frac{1}{2}}a\|^2 + \beta - \frac{1}{4}a^T Q^{-1}a$

化为 SOCP: $\min u_0$

s.t. $\bar{u} = Q^{\frac{1}{2}}x + \frac{1}{2}Q^{-\frac{1}{2}}a, Ax = b, x \geq 0, \begin{pmatrix} u_0 \\ \bar{u} \end{pmatrix} \succeq_Q 0$

一般地 $q(x) = x^T B^T B x + a^T x + \beta \leq 0$

$\Leftrightarrow q(x) = \|Bx\|^2 + a^T x + \beta \leq 0$

$\Leftrightarrow \begin{bmatrix} -a^T x - \beta \\ 1 \\ Bx \end{bmatrix} \in Q_r \Rightarrow \begin{bmatrix} \frac{-a^T x - \beta - 1}{2} \\ Bx \end{bmatrix} \preceq \frac{1 - a^T x - \beta}{2}$

Robust LP: $\min c^T x$ s.t. $a_i^T x \leq b_i$ (! Uncertainty in a_i, b_i, c)

- deterministic model: constraints hold for every $a_i \in \mathcal{E}_i$
- stochastic model: constraints hold with $\Pr \eta$ i.e. $P(a_i^T x \leq b_i) \geq \eta$

Deterministic: $\mathcal{E}_i = \{ \bar{a}_i + P_i u \mid \|u\|_2 \leq 1 \}$

Robust LP: $\min c^T x$ s.t. $a_i^T x \leq b_i$ for all $a_i \in \mathcal{E}_i$

$\Leftrightarrow \min c^T x$ s.t. $\bar{a}_i^T x + \|P_i^T x\|_2 \leq b_i \leftarrow \text{SOCP}$

Stochastic:

$a_i \sim \mathcal{N}(\bar{a}_i, \Sigma_i)$ then $P(a_i^T x \leq b_i) = \Phi\left(\frac{b_i - \bar{a}_i^T x}{\|\Sigma_i^{\frac{1}{2}} x\|_2}\right)$

Robust LP: $\min c^T x$ s.t. $\bar{a}_i^T x + \Phi^{-1}(\eta) \|\Sigma_i^{\frac{1}{2}} x\|_2 \leq b_i \leftarrow \text{SOCP}$

四. 半定规划 (SDP)

$$\text{LMI: } \min C^T x \quad \text{s.t.} \quad x_1 A_1 + x_2 A_2 + \dots + x_n A_n + B \preceq 0, \quad Gx = h$$

其中 $C \in \mathbb{R}^n$, $A_i \in S^m$, $i=1,2,\dots,n$, $B \in S^m$, $G \in \mathbb{R}^{p \times n}$, $h \in \mathbb{R}^p$ 比较类似线性规划.

取 $X = \sum_{i=1}^n x_i A_i + B$ $\leadsto$

标准形式: $\min \langle C, X \rangle \quad \text{s.t.} \quad \langle A_1, X \rangle = b_1, \dots, \langle A_m, X \rangle = b_m, \quad X \succeq 0$ ($\langle A, X \rangle = \text{tr}(A^T X)$)

对偶形式: $\min -b^T y \quad \text{s.t.} \quad C - \sum_{i=1}^m y_i A_i \succeq 0$

转化方式: Lagrangian: $\mathcal{L}(X, y) = \langle C, X \rangle + \sum_i y_i (b_i - \langle A_i, X \rangle) = b^T y + \langle C - \sum_{i=1}^m y_i A_i, X \rangle$

记 $g(y) = \inf_{X \succeq 0} \mathcal{L}(X, y)$, 若 $C - \sum_{i=1}^m y_i A_i \succeq 0$, $g(y) = b^T y$; 否则 $g(y) = -\infty$

故最大化 $g(y) = b^T y$ 即可

$$\text{LP} \subseteq \text{SDP}: \quad \text{LP}: \min c^T x \quad \text{s.t.} \quad Ax \leq b \quad \quad \text{SDP}: \min c^T x \quad \text{s.t.} \quad \text{diag}(Ax-b) \leq 0$$

$$\text{SOCP} \subseteq \text{SDP}: \quad \text{SOCP}: \min f^T x \quad \text{s.t.} \quad \|A_i x + b_i\|_2 \leq c_i^T x + d_i, \quad i=1 \dots m$$

$$\text{SDP}: \min f^T x \quad \text{s.t.} \quad \begin{bmatrix} (c_i^T x + d_i)I & A_i x + b_i \\ (A_i x + b_i)^T & c_i^T x + d_i \end{bmatrix} \succeq 0 \quad i=1 \dots m$$

二次约束二次规划问题半定规划松弛.

$$\min_{x \in \mathbb{R}^n} x^T A_0 x + 2b_0^T x + c_0 \quad \text{s.t.} \quad x^T A_i x + 2b_i^T x + c_i \leq 0 \quad i=1 \dots m$$

半定规划松弛: 利用 $x^T A x = \text{tr}(x^T A x) = \text{tr}(A x x^T) = \langle A, x x^T \rangle$

$$\text{故只需} \quad \min_{x \in \mathbb{R}^n} \langle A_0, X \rangle + 2b_0^T x + c_0 \quad \text{s.t.} \quad \langle A_i, X \rangle + 2b_i^T x + c_i \leq 0 \quad i=1 \dots m, \quad X = x x^T$$

$$\text{进一步} \quad x^T A x + 2b^T x + c = \left\langle \begin{pmatrix} A & b \\ b^T & c \end{pmatrix}, \begin{pmatrix} X & x \\ x^T & 1 \end{pmatrix} \right\rangle \stackrel{\text{def}}{=} \langle \bar{A}, \bar{X} \rangle$$

唯一非线性

部分
↓
替为 $X \succeq x x^T$

$$\Leftrightarrow \bar{X} \succeq 0$$

$$\text{Final:} \quad \min \langle \bar{A}_0, \bar{X} \rangle, \quad \text{s.t.} \quad \langle \bar{A}_i, \bar{X} \rangle \leq 0, \quad \bar{X} \succeq 0, \quad \bar{X}_{n+1, n+1} = 1$$

最大割的半定松弛问题:

$G = (V, E, w_{ij} = w_{ji} \geq 0)$ 找 V 的一个分割 $V = S \cup \bar{S}$, 使 S 与 $\bar{S}$ 间连边和最大

记 $x_i = 1, i \in S; x_i = -1, i \in \bar{S}$, 则 Hope: $\max \sum_{i < j} \frac{1}{2}(1 - x_i x_j) w_{ij} \quad (*)$

令 $W = (w_{ij}) \in S^n$, $C = -\frac{1}{4} (\text{Diag}(W\mathbf{1}) - W)$

即 $(*) \Leftrightarrow \min x^T C x \quad \text{s.t.} \quad x_i^2 = 1 \quad i = 1, \dots, n$

松弛: $\min \langle C, X \rangle, \text{ s.t. } X_{ii} = 1 \quad i = 1, \dots, n, \quad X \geq 0 \quad \text{rank}(X) = 1$ 去掉即为松弛

极小化最大特征值: $\min_x \lambda_{\max}(A_0 + \sum_i x_i A_i)$ 利用 $\lambda_{\max}(M) \leq t \Leftrightarrow A \preceq tI$

SDP: $\min z \quad \text{s.t.} \quad zI - \sum_i x_i A_i \succeq 0$

极小化二范数问题: $A(x) = A_0 + \sum_i x_i A_i$, $\min_x \|A(x)\|_2$

SDP: $\min_{x,t} t \quad \begin{pmatrix} tI & A(x) \\ A(x) & tI \end{pmatrix} \succeq 0$

Schur 补: $U = \begin{pmatrix} A & B \\ B^T & C \end{pmatrix}$ A, C 对称, $A \succ 0$; U 半正定 $\Leftrightarrow C - B^T A^{-1} B \succeq 0$